

Computer Basics: Outline

- Hardware and Memory
- Programs
- Programming Languages and Compilers
- Java Byte-Code
- Graphics Supplement

Hardware and Software

- Computer systems consist of *hardware* and *software*.
 - Hardware includes the *tangible* parts of computer systems.
 - Software includes *programs* - sets of instructions for the computer to follow.
- Familiarity with hardware basics helps us understand software.

Hardware and Memory

- Most modern computers have similar components including
 - Input devices (keyboard, mouse, etc.)
 - Output devices (display screen, printer, etc.)
 - A processor
 - Two kinds of memory (main memory and auxiliary memory).

The Processor

- Also called the *CPU* (central processing unit) or the *chip* (e.g. Pentium processor)
- The processor **processes** a program's instructions.
- It can process only very simple instructions.
- The power of computing comes from speed and program intricacy.

Memory

- Memory holds
 - programs
 - data for the computer to process
 - the results of intermediate processing.
- Two kinds of memory
 - main memory
 - auxiliary memory

Main memory

- Working memory used to store
 - The current program
 - The data the program is using
 - The results of intermediate calculations
- Usually measured in megabytes (e.g. 8 gigabytes of RAM)
 - RAM is short for *random access memory*
 - A *byte* is a quantity of memory

Auxiliary Memory

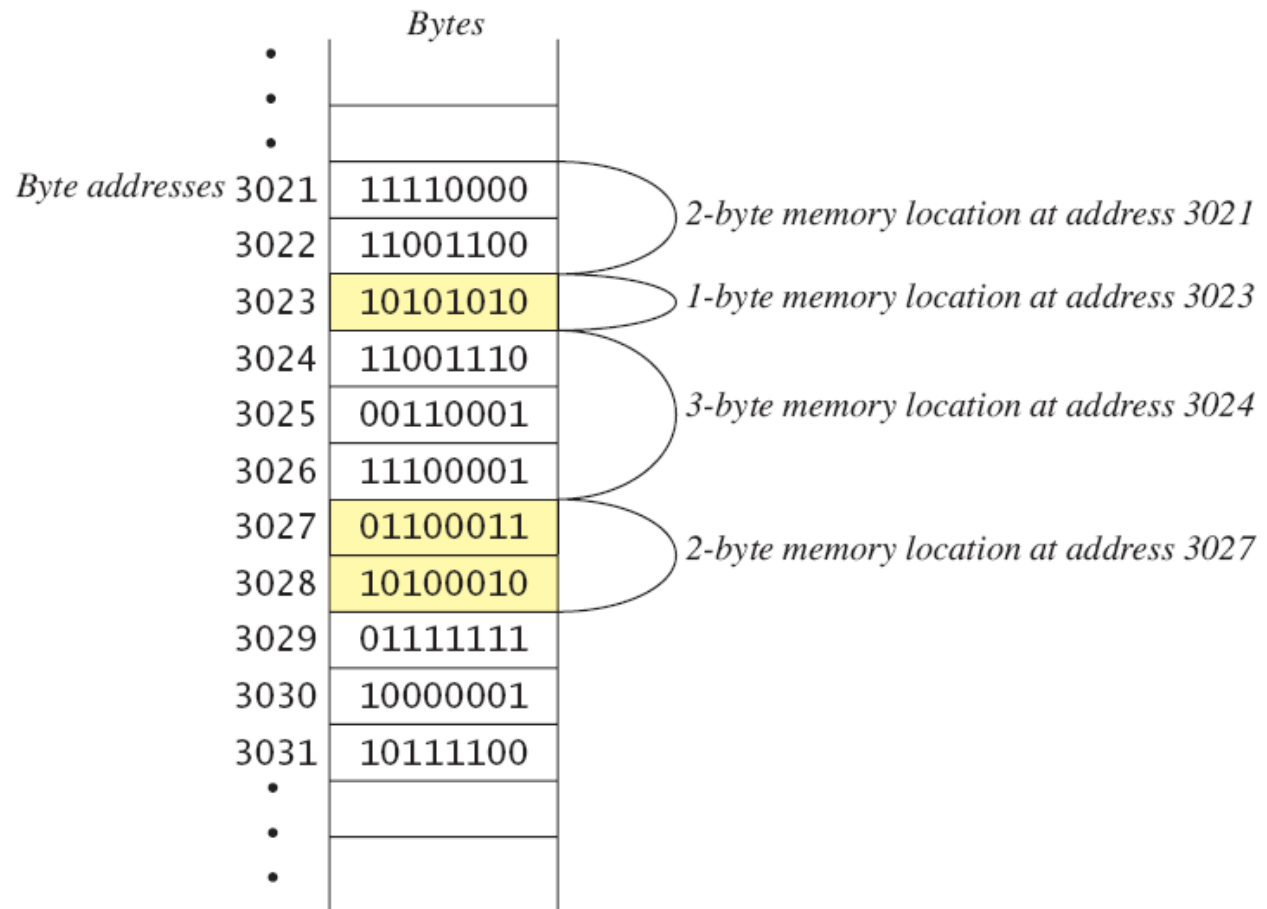
- Also called *secondary memory*
- Disk drives, CDs, DVDs, flash drives, etc.
- More or less permanent (nonvolatile)
- Usually measured in gigabytes (e.g. 50 gigabyte hard drive)

Bits, Bytes, and Addresses

- A *bit* is a digit with a value of either 0 or 1.
- A *byte* consists of 8 bits.
- Each byte in main memory resides at a numbered location called its *address*.

Main Memory

- Figure 1.1



Storing Data

- Data of all kinds (numbers, letters, strings of characters, audio, video, even programs) are encoded and stored using 1s and 0s.
- When more than a single byte is needed, several adjacent bytes are used.
 - The address of the first byte is the address of the unit of bytes.

Files

- Large groups of bytes in auxiliary memory are called *files*.
- Files have names.
- Files are organized into groups called *directories* or *folders*.
- Java programs are stored in files.
- Programs files are copied from auxiliary memory to main memory in order to be run.

0s and 1s

- Machines with only 2 stable states are easy to make, but programming using only 0s and 1s is difficult.
- Fortunately, the conversion of numbers, letters, strings of characters, audio, video, and programs is done automatically.

Programs

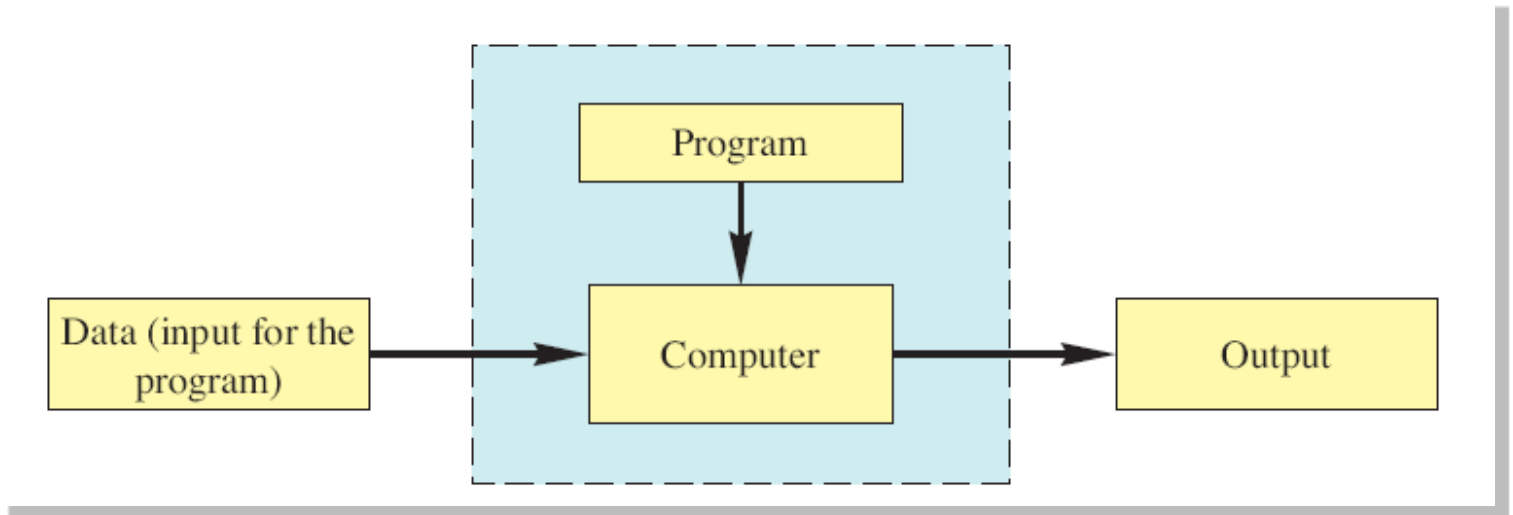
- A *program* is a set of instructions for a computer to follow.
- We use programs almost daily (email, word processors, video games, bank ATMs, etc.).
- Following the instructions is called *running* or *executing* the program.

Input and Output

- Normally, a computer receives two kinds of input:
 - The program
 - The *data* needed by the program.
- The output is the result(s) produced by following the instructions in the program.

Running a Program

- Figure 1.2



- Sometimes the computer and the program are considered to be one unit.
 - Programmers typically find this view to be more convenient.

The Operating System

- The *operating system* is a supervisory program that oversees the operation of the computer.
- The operating system retrieves and starts program for you.
- Well-known operating systems including: Microsoft Windows, Apple's Mac OS, Linux, and UNIX.

Programming Languages

- *High-level languages* are relatively easy to use
 - Java, C#, C++, Visual Basic, Python, Ruby.
- Unfortunately, computer hardware does not understand high-level languages.
 - Therefore, a high-level language program must be translated into a *low-level language*.

Compilers

- A *compiler* translates a program from a high-level language to a low-level language the computer can run.
- You *compile* a program by running the compiler on the high-level-language version of the program called the *source program*.
- Compilers produce *machine-* or *assembly-language* programs called *object programs*.

Compilers

- Most high-level languages need a different compiler for each type of computer and for each operating system.
- Most compilers are very large programs that are expensive to produce.